



A comprehensive trustworthy framework for connected machine learning and secure interconnected AI solutions  
GA ID: 101168560

# Amendment request Letter

In what follows, we provide the justification and an analysis of the requested amendment changes. The changes in Part A and Part B are summarized in the following tables.

Table 1: Summary of changes appearing in Part A

| Changes                                                                                                                                                                                                                                                                                                                           | Justification                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IMT (PIC: 999849326) has withdrawn and is replaced by the University of Patras (UPatras, PIC: 999894528)                                                                                                                                                                                                                          | Transition of the Principal Investigator from IMT to UPatras                                                                                                                                                                          |
| University of Patras (UPatras, PIC: 999894528) has been added into the consortium                                                                                                                                                                                                                                                 | Transition of the Principal Investigator from IMT to UPatras                                                                                                                                                                          |
| BEVD (PIC: 879892724) has been added into the consortium                                                                                                                                                                                                                                                                          | BEVD's expertise in technology transfer, business sustainability strategies, and requirements engineering. BEVD personnel is a strategic collaborator of AVL and will complement the efforts toward achieving the project objectives. |
| The list of participants for tasks T1.1, T2.1, T2.4 and T3.2 has been updated                                                                                                                                                                                                                                                     | To include BEVD's involvement in these tasks.                                                                                                                                                                                         |
| Intra-work package resource distribution adjustments: <ul style="list-style-type: none"> <li>• UNIP: 9 PMs from T5.1 and another 9 PMs from T5.4 have been shifted to T5.2 and T5.5 respectively</li> <li>• UNICA: 3 PMs have been shifted from T4.5 to T4.1</li> <li>• AVI: 3 PMs have been shifted from T4.1 to T4.2</li> </ul> | Adjustments to reflect the use case requirements focus                                                                                                                                                                                |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <p>The Financial Information has been updated:</p> <ul style="list-style-type: none"> <li>• SAP - prevPersonnel: 352.000 €, newPersonnel: 0 €, prevOtherDirect: 38.600 €, newOtherDirect: 0 €</li> <li>• ICCS - prevPersonnel: 345.000 €, newPersonnel: 410.000 €, prevOtherDirect: 29.000 €, newOtherDirect: 48.866 €</li> <li>• ISI - prevPersonnel: 325.000 €, newPersonnel: 420.000 €, prevOtherDirect: 30.000 €, newOtherDirect: 42.867 €</li> <li>• AEGIS - prevPersonnel: 318.200 €, newPersonnel: 503.200 €, prevOtherDirect: 25.000 €, newOtherDirect: 37.867 €</li> <li>• IMT - prevPersonnel: 221.000 €, newPersonnel: 0 €, prevOtherDirect: 20.000 €, newOtherDirect: 0 €</li> <li>• UPatras - prevPersonnel: 0 €, newPersonnel: 220.000 €, prevOtherDirect: 0 €, newOtherDirect: 21.000 €</li> <li>• AVL - prevPersonnel: 288.600 €, newPersonnel: 229.400 €, prevOtherDirect: 18.000 €, newOtherDirect: 18.600 €</li> <li>• BEVD - prevPersonnel: 0 €, newPersonnel: 48.600 €, prevOtherDirect: 0 €, newOtherDirect: 10.000 €</li> </ul> | <p>Reflects new resource distribution: resources transferred from AVL to BEVD, IMT to UPatras and SAP to ICCS, ISI, AEGIS</p>           |
| <p>Updates in time planning (Gantt):<br/>The duration of T6.3 in the task description and Gantt chart has been revised from M24-M30 to M21-M34, while the duration of T6.1 has been adjusted from M19-M33 to M13-M33</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Fixed inconsistencies between task descriptions and Gantt chart</p>                                                                  |
| <p>Risk 1 description correction</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>References to the UK's participation in H2020 have been removed, as this detail is unnecessary.</p>                                  |
| <p>Risk 10 description correction</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Removed the reference to the "partner roles (Innovation Manager and DEM)" which are not defined in the technical annex.</p>          |
| <p>Increase of overall project effort by 24 PMs (from 704 to 728)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>An additional 24 PMs were added to support the partners (ICCS, ISI, AEGIS, UPatras and BEVD) undertaking the tasks that are left</p> |

|  |                       |
|--|-----------------------|
|  | from SAP, IMT and AVL |
|--|-----------------------|

*Table 2: Summary of changes appearing in Part B*

| Changes                                                                                                                | Justification                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Replaced references to “IMT” with “UPatras”                                                                            | UPatras takes over all IMT tasks and responsibilities and appropriate references were added across the text        |
| Introduced “BEVD” in the text                                                                                          | BEVD undertakes specific roles and responsibilities from AVL and appropriate references were added across the text |
| Updated figures and content in Section 3.2 to reflect the new proposed budget distribution and the changes Pages 40-41 | The changes reflect the addition of BEVD and the replacement of IMT with UPatras                                   |
| Modifications in the Assets table (1.1.2.4 Technologies Introduced to CoEvolution and TRL Levels) – Page 11            | UPatras brings in baseline technology to replace the one that IMT was bringing                                     |
| Replacement of “deep unrolling” with “model-based AI” – Pages 11, 19 & 21                                              | Broaden the scope of the research and update it with a more modern technology.                                     |
| Modification of Pilot 2: Securing the AI Supply Chain in the Autonomous Vehicles – Page 27                             | Provide further details to the Pilot.                                                                              |
| Fine-tuning of KPIs                                                                                                    | Fixes to improve clarity and precision.                                                                            |

In more detail, the amendment request concerns the following changes:

1. Addition of 2 new beneficiaries
2. Beneficiary termination
3. Change of Annex 1 - description of the action
4. Change of Annex 2 - estimation of the budget

These change(s) are necessary for the following reason(s)

## 1.1 Consortium composition and subsequent changes

Following a formal request from partner 4 (IMT, PIC: 999849326), IMT has withdrawn and is replaced by the University of Patras (UPatras, PIC: 999894528) due to the transition of the Principal Investigator. Additionally, BEVD (PIC: 879892724) has been introduced as a new partner to take on responsibilities primarily in *Task 2.4 Exploitation activities and long-term sustainability* and *Task 3.2 Use-Case specifications and Requirements*.

This decision is based on BEVD's expertise in technology transfer, business sustainability strategies, and requirements engineering. BEVD personnel is a strategic collaborator of AVL and will complement the efforts toward achieving the project objectives.

Supporting documents on the agreement from all partners have been provided, reflecting the withdrawal of IMT, its replacement with UPatras, and the addition of a new partner, BEVD.

The figures and content in Section 3.2 have been updated to reflect the new proposed budget distribution and the changes following the addition of a new partner, namely BEVD to undertake some tasks instead of AVL, and the withdrawal of IMT and its subsequent replacement with UPatras.

The Assets table (*Section 1.1.2.4 Technologies Introduced to CoEvolution and TRL Levels*) has also been modified. Originally, "PlanIoT" was mentioned as an asset to be used by the project. However, this work is IMT's IP and with their departure from the project it has to be replaced. UPatras is bringing ComDeX, a Context-aware Federated Platform for Data Exchange which provides equivalent functionality to be used by CoEvolution. ComDeX extends PlanIoT functionality and considers efficient planning of IoT data exchange between Federated Data spaces, as well as performing collaborative Machine Learning (e.g. Federated Learning) between such data spaces using context information. In Coevolution such a context-aware mechanism is part of *Task 4.4 AI Model Context-Awareness Enablers for Security, Robustness and Performance*.

## 1.2 Methodology changes

In terms of methodology, the consortium would like to introduce some clarifications that will facilitate the execution of the project. First, the term "deep unrolling" has been replaced with "model-based AI". "Model-based AI" is an approach in AI where models are explicitly designed based on prior knowledge, principles, or constraints from the domain, offering better interpretability, robustness, and efficiency compared to purely data-driven methods. Deep unrolling is a specific technique within this framework, where iterative optimization algorithms are reformulated as neural networks, combining theoretical foundations with learnable parameters. Thus, this change broadens the scope to include a wider range of methods, emphasizing flexibility and alignment with the project's overarching goals.

Further modifications have been made to *Pilot 2 Securing the AI Supply Chain in the Autonomous Vehicles*. Specifically, this specific pilot has been divided into sub-use cases. Since the overall pilot architecture includes both perception and control, we aim to study in more detail each of these parts and therefore we are introducing an intermediary phase where each of them is examined

individually. This approach will facilitate more efficient monitoring of these two components and allow for their smoother integration.

The project's KPIs have been fine-tuned for clarity and precision. KPI-1.2, initially set as *"Adversarial Attack Detection Rate  $\geq 90\%$ "* has been revised to *"System accuracy degradation under attack  $\leq 10\%$ "*. This adjustment aims to provide a more precise and measurable metric.

It aligns better with machine learning practices, where the impact of adversarial attacks on overall system performance is a key evaluation criterion. The revised KPI ensures a clearer and more accurate assessment of the system's robustness under attack. KPI-3.1, initially set as *"Design and Develop at least 2 variants of AI MBOM structures"* has been revised to *"Design and Develop at least 2 variant instances of the AI MBOM"* emphasizing the development of functional instances of the AI MBOM, providing a more precise and actionable metric for evaluation. Pilot 2 - KPI-UC2.1 initially set as *"Efficiency in mitigating the attack in terms of the CAV's geo-localization"* has been revised to *"Efficiency in mitigating the attack in terms of the CAV's perception engine e.g., geo-localization, multi-object detection and tracking"* in order to broaden the scope of the KPI to reflect a more comprehensive evaluation of the CAV's perception engine. By including aspects such as geo-localization, multi-object detection, and tracking, the updated KPI ensures a more holistic assessment of the system's efficiency in mitigating attacks, aligning with the objectives of Pilot 2.

## 1.3 Impact section changes

In terms of impact, the consortium would like to suggest some changes for the clarification of certain concepts and the better delineation of the goals.

Adjustments have also been made to impact-related metrics.

Expected Outcome #1 has been revised to focus on system accuracy degradation under adversarial attacks rather than detection rates (initially set as *"Adversarial Attack Detection Rate by the AI models enhanced with CoEvolution gadgets and security self-awareness  $> 95\%$ "* has been revised to *"System accuracy degradation under adversarial attack on AI models enhanced with CoEvolution gadgets and security self-awareness  $\leq 10\%$ "*, following the same approach as KPI-1.2). Destination Impact #1 has been removed, as its revised metric is already reflected in KPI-1.2. Destination Impact #5 now specifies the adoption rate of AI Model Bill of Materials among pilot stakeholders, ensuring a more measurable outcome (initially set as *"Adoption of AI Model Bill of Materials (MBOM)"* has been revised to *"Adoption of AI Model Bill of Materials from pilot stakeholders  $\geq 80\%$ "*).

Similarly, Destination Impact #6 now defines the adoption of CoEvolution within the last year of the project, making the target more specific and time-bound (initially set as “*Adoption rate of CoEvolution = 100%*” has been revised to “Adoption rate of CoEvolution within the last year of the project = 100%”). Further refinements have been made to KPIs related to the project’s Economic and Technological Outcomes (*Section 2.1.3*). The previous KPI, “Attack Detection Rate  $\geq 95\%$ ” has been removed. Instead, the project will assess “*System accuracy degradation under attack  $\leq 10\%$* ”, which has already been defined as KPI-1.2. Finally, adjustments have been made to the KPI of *Table 2.1: Dissemination Plan – Measures, Means, Target Groups, and Target Values (Standardisation)*. The previous requirement of “*Number of standardization contributions to different SDOs with adoption status:  $\geq 4$* ” has been revised to “Number of standardization contributions to different SDOs:  $\geq 4$ ”. The revision removes the adoption status requirement from the KPI to reflect the reality that achieving adoption by SDOs typically requires more time than the three-year timeframe of the project. While this target is ambitious, the project remains committed to pursuing adoption to the best of its ability.

### 1.3.1 Update tasks

The list of participants for tasks T1.1, T2.1, T2.4 and T3.2 has been updated to include BEVD’s involvement in these tasks.

### 1.3.2 Intra-work package resource distribution has been fine-tuned

The person-month (PM) allocations have been adjusted as follows: For UNIPI, PMs have been reallocated from T5.1 and T5.4 to T5.2 and T5.5. Additionally, for UNICA, 3 PMs have been shifted from T4.5 to T4.1, while for AVI, 3 PMs have been shifted from T4.1 to T4.2.

### 1.3.3 Update Annex 2-estimated budget

The proposed changes result in a subsequent increase of the overall effort by 24 PMs. These changes are explained in the context of:

- The withdrawal of partner “SAP”, (PIC: 999911212) and the undertaking of its tasks from participating partners ICCS, ISI and AEGIS
- The withdrawal of partner “IMT” (PIC: 999849326) and the undertaking of its tasks from the new partner “UPATRAS” (PIC: 999894528)
- The undertaking of part of the tasks of partner “AVL” (PIC: 999952243) from the new partner BEVD (PIC: 879892724)

Some changes required additional effort from what was originally considered resulting in an increase of the overall project effort by 24PMs (728PMs from 704PMs). The following table provides a summary of the changes with a short explanation. The first column groups the changes in the PMs based on the relevant event (substitution of partners assigned tasks). The rest of the columns provide details about the change in the PMs for each involved partners. A more detailed justification for the changes is provided in the text following the table:

| Event                                          | Partner concerned | Initial PM allocation | Proposed PM allocation | Difference | Change log/explanation                                                                                                                                                  |
|------------------------------------------------|-------------------|-----------------------|------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Substitution of SAP from ICCS, ISI, AEGIS      | SAP               | 44                    | 0                      | -44        | Withdrawal from project and all involved WPs (WP1:1->0PM, WP2:7->0PM, WP3:7->0PM, WP4:24->0PM, WP6:5->0PM)                                                              |
|                                                | ICCS              | 69                    | 82                     | 13         | 13PMs moved from SAP tasks to ICCS for the same tasks (WP1:15->16PM, WP3:2->8PM, WP4:9->15PMs)                                                                          |
|                                                | ISI               | 65                    | 84                     | 19         | 19PMs moved from SAP tasks to ISI for the same tasks (WP2:1->2PM, WP4:10->28PM)                                                                                         |
|                                                | AEGIS             | 43                    | 68                     | 25         | 12PMs moved from SAP tasks to AEGIS for the same tasks, plus an additional 13PMs to support the same tasks (WP2:8->21PM, WP3:1->5PMs, WP6:10->18PM)                     |
| Substitution of IMT from UPATRAS               | IMT               | 34                    | 0                      | -34        | Withdrawal from project and all involved WPs (WP1:1->0PM, WP2:4->0PM, WP3:3->0PM, WP4:16->0PM, WP6:10->0PM)                                                             |
|                                                | UPATRAS           | 0                     | 44                     | 44         | 33PMs moved from IMT tasks to UPATRAS for the same tasks plus an additional 10PMs to support the same tasks (WP1:0->1PM, WP2:0->4PM, WP3:0->7PM, WP4:0->20PM, WP6:12PM) |
| Substitution part of the work of AVL from BEVD | AVL               | 39                    | 31                     | -8         | cross-WP redistribution of PMs (WP7->WP1:+1PM, WP7->WP2:+3PM) and subsequent withdrawal of 8PMs from some tasks (WP1:2->1, WP2:15->10PM, WP3:6->4PM, WP6:10->10PM)      |
|                                                | BEVD              | 0                     | 9                      | 9          | 8PMs moved from AVL tasks to BEVD for the same tasks, plus an additional 1PM to support the same tasks (WP1:0->1PM, WP2:0->5PM, WP3:0->3PM)                             |
| <b>Sums</b>                                    |                   | <b>294</b>            | <b>318</b>             | <b>24</b>  |                                                                                                                                                                         |

The table is explained in what follows:

#### **SAP to ICCS, ISI, AEGIS**

Following a formal request from Partner 9 (SAP, PIC: 999911212), SAP has withdrawn from the project, and its allocated resources (44 PMs in total) have been reassigned to ICCS, ISI, and AEGIS.

More specifically:

- ICCS: direct shift of 13 PMs from SAP in T1.1, T3.1, T3.3, T4.2 (initial resources: 69, final resources: 82)
- ISI: direct shift of 19 PMs from SAP in T2.2, T4.1, T4.2, T4.5 (initial resources: 65, final resources: 84)
- AEGIS: direct shift of 12 PMs from SAP in T2.1, T2.4, T2.5, T3.2, T6.1, T6.3

In comparison to the original PM allocation, AEGIS has been assigned additional PMs for specific tasks, in addition to the direct transfer of resources from SAP.

More specifically: For T2.1, the PMs have increased from 8 to 12 (direct shift of 2 PMs from SAP); for T2.4, the PMs have increased from 0 to 6 (direct shift of 3 PMs from SAP); for T2.5, the PMs have increased from 0 to 3 (direct shift of 1 PM from SAP); for T3.2, the PMs have been increased from 1 to 5 (direct shift of 1 PM from SAP); for T6.1, the PMs have been increased from 8 to 13 (direct shift of 3 PMs from SAP) and for Task T6.3, the PMs have increased from 2 to 5 (direct shift of 2 PMs from SAP).

The justification for these changes is to ensure a closer alignment with the evolving needs of the project. Specifically, in WP2, AEGIS has been allocated additional resources to enhance the project's efforts in communication, dissemination, and exploitation, with a particular focus on reaching a wider range of stakeholder groups and communities of interest. In WP3, the increase in resources supports the extension of the work package to include an analysis of legal and ethical requirements. Finally, in WP6, due to the scale of the company, AEGIS requires additional effort to contribute effectively to the integration, testing, pilot deployment, and demonstration of the project's framework.

#### **IMT to UPATRAS**

In comparison to the original PM allocation to IMT, UPatras has been assigned additional PMs for specific tasks as follows: For Task T3.1, the PMs have increased from 1 to 3; for T3.3, the PMs have increased from 2 to 4; for Task T4.4, the PMs have increased from 8 to 12; and for Task T6.2, the PMs have increased from 6 to 8. The justification for these changes is to better align with the project's evolving needs and to ensure adequate resource allocation for the successful completion of these tasks. Additionally, these changes have been reflected in Annex 2 - Estimated Budget for the Action. Due to differences in the PM rate of the two institutions (IMT and Upatras)



and the addition of 1000 euros for travel costs to the originally allocated 20.000 euros the budget of UPatras remains the same with the originally allocated to IMT.

More specifically:

T3.1: UPatras requires two additional PMs to further enhance state-of-the-art analysis, particularly in networked AI systems—a key research area in CoEvolution. Designing security-by-design approaches for these systems is complex, requiring the exploration of existing methods. For instance, attack graphs and trust computation are commonly used to mitigate cyber attack risks in networked systems.

T3.3: UPatras requires two additional PMs to further contribute in defining the CoEvolution reference architecture. CoEvolution's architecture requires defining context-aware information flows as well as controlling flows between the different layers. UPatras' experience related to enabling context-aware collaborative AI in networked systems will be valuable for this task.

T4.4: UPatras requires four additional PMs to further develop the technological framework for context-aware AI resilience in distributed FL and decentralized P2PL architectures. Specifically, it will enable context-aware data flows by modeling raw data and metadata as property graph entities using FIWARE data models—an approach in which UPatras has significant experience. These data flows and actions will be disseminated across networked AI systems via a publish/subscribe software overlay enabling distributed FL and decentralized P2PL architectures. Finally, leveraging self-adaptation techniques, a decision-support agent will dynamically adjust its policies based on the context disseminated within the overlay.

T6.2: UPatras requires two additional PMs to further enhance the development of the dataset simulator. UPatras will exploit past experience in generating datasets for smart spaces and extend it for generating trusted data for the training phase of CoEvolution AI models. Existing datasets and virtual environments (related to pilot cases) will be used as input to the simulator to extract patterns/relationships and generate trustworthy datasets.

### **AVL to BEVD**

BEVD has been introduced as a new partner to take on key responsibilities, primarily in Task 2.4: "Exploitation Activities and Long-Term Sustainability" and Task 3.2: "Use-Case Specifications and Requirements", due to its proven expertise in technology transfer, business sustainability strategies, and requirements engineering. BEVD personnel serve as strategic collaborators with AVL and will complement the automotive domain contributions toward achieving the project objectives related to these tasks.

- T1.1: BEVD is assigned 1 PM to support managerial tasks, in line with all other involved partners.
- T2.1: BEVD is assigned 1 PM to contribute to dissemination and communication activities, similar to all partners.

- T2.4: BEVD is assigned 4 PMs to take over leadership of the exploitation task and strengthen the technology transfer efforts towards the automotive industry, which accounts for two-thirds of the project's use cases.
- T3.2: BEVD is assigned 3 PMs to lead the task and facilitate the interface between end users and technology providers, with a strong focus on the automotive domain.

These resources are secured through the reallocation of 2 PMs from each of the following AVL tasks: T2.4, T3.2, T6.1, and T6.3. Aside from the transfer of leadership from AVL to BEVD in T2.4 and T3.2, the remaining resources have been freed from WP6, as the use cases between AVL and AVI have become more compatible. This shift is attributed to AVI's recent purchase of a research testbed, which reduces the need for extensive integration and interfacing work.

The purchase costs allocated to BEVD (EUR 10,000) are entirely related to travel and subsistence. No costs are allocated to purchasing goods, subcontracting, or external services. Additionally, although the travel and subsistence costs exceed 15% of personnel costs, they are necessary to fulfil BEVD's role in coordinating industrial partner activities, requirements collection, and exploitation work. BEVD is expected to attend at least six project meetings, two review meetings, and participate in two dissemination activities, including two conference presentations as first author. The travel budget, based on an estimated EUR 1,000 per trip, covers a minimum of 10 trips. Compared to other partners, BEVD's overall travel and subsistence budget is significantly lower, reflecting a smaller workforce, while ensuring necessary participation in key project activities.

#### 1.3.4 Updates in time planning for Tasks and Milestones

Additionally, the duration of T6.3 in the task description and Gantt chart has been revised from M24-M30 to M21-M34, while the duration of T6.1 has been adjusted from M19-M33 to M13-M33 to enhance work coordination and ensure more efficient project execution.

#### 1.3.5 Risk management

Risk 1 has been updated to remove references to the UK's participation in H2020, as this detail is unnecessary.

These changes aim to improve work coordination and ensure better alignment of project activities.

Please let us know if you need any further information on this matter.

Sincerely yours,  
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